ONCOLOGY · NEET-SS / FET / INI-SS

Metastatic bone disease

1. Epidemiology

- **Metastasis is the commonest malignant lesion of bone** (far commoner than primary bone sarcoma); in a patient **over 40 with a destructive bone lesion, metastasis (or myeloma) is the most likely diagnosis**.
- **Bone is the third commonest site of metastasis** (after lung and liver).
- **Primary tumours that commonly metastasise to bone: breast, prostate, lung, kidney, thyroid**; **breast, prostate and lung account for about 80%** of skeletal metastases.
  - Breast and prostate: bone metastases in up to **70%** of advanced disease.
  - **Commonest primary in women: breast**; **in men: prostate** (lung close behind).
- **Commonest site of bone metastasis: spine** (**thoracic about 70%**, lumbar about 20%, cervical about 10%), then **pelvis, ribs, proximal femur** (the **commonest long bone**), skull and **proximal humerus**. Metastases are commonest in **red-marrow-rich axial and proximal appendicular bones**.
- **Acrometastases** (distal to elbow and knee: hand and foot) are rare; the **commonest primary is lung**.
- **Children: neuroblastoma** (commonest under 5), Wilms tumour (rare; clear cell sarcoma of kidney is the bone-seeking renal tumour), rhabdomyosarcoma, retinoblastoma.

2. Pathophysiology

- **Routes:** haematogenous (main), via **Batson's valveless vertebral venous plexus** (explains spread of **prostate and breast** cancer to the spine and pelvis when intra-abdominal or intrathoracic pressure rises), direct extension (rare).
- **"Seed and soil" hypothesis (Paget):** marrow provides a favourable microenvironment (growth factors, adhesion molecules, chemokines such as CXCL12/CXCR4).
- **The vicious cycle of osteolysis:** tumour cells secrete **PTHrP** (and IL-6, IL-11), which stimulates osteoblasts to express **RANKL** (and reduces OPG); RANKL activates **osteoclasts**; bone resorption releases **TGF-beta, IGF and calcium** from the matrix, which stimulate further tumour growth. Basis for **bisphosphonates and denosumab** (anti-RANKL antibody).
- **Osteoblastic metastases (prostate):** tumour cells secrete **endothelin-1**, Wnt proteins and BMPs, stimulating osteoblasts; the new bone is woven and mechanically weak.
- A lesion must destroy about **30-50% of the bone mineral** before it is visible on radiographs.

Radiological pattern by primary

Pattern	Primaries
Purely lytic	**Kidney (renal cell)**, **thyroid**, lung (non-small cell, most), melanoma, GI tract, adrenal, myeloma
Osteoblastic (sclerotic)	**Prostate** (classic), carcinoid, small cell lung cancer, medulloblastoma, bladder (occasionally), Hodgkin lymphoma, treated breast cancer
Mixed	**Breast**, lung, cervix, ovary, testicular
Expansile, "blow-out" lytic	Renal and thyroid

- **Renal cell and thyroid metastases are hypervascular**: **preoperative embolisation** reduces blood loss (surgery within about 24-72 hours of embolisation); do not do an unplanned open biopsy or curettage without being ready for major bleeding.

3. Clinical features

- **Pain**: the commonest symptom; initially **night and rest pain** (biological), later **mechanical/functional pain on weight bearing** (impending fracture).
- **Pathological fracture**: first presentation in some patients.
- **Hypercalcaemia**: the **commonest metabolic complication** (up to 10-30% of patients with advanced cancer; breast, lung, myeloma; renal through PTHrP): thirst, polyuria, confusion, constipation, nausea, arrhythmias. Treat with **IV fluids, IV bisphosphonate (zoledronic acid)**, **denosumab** if refractory or renal impairment, calcitonin for rapid short-term effect, steroids in lymphoma/myeloma.
- **Metastatic spinal cord compression (MSCC)**: back pain (often thoracic, band-like, worse lying down or with straining) precedes neurological deficit; weakness, sensory level, sphincter disturbance late.
- Marrow failure (anaemia), constitutional symptoms.
- **Skeletal-related events (SREs)**: pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone, hypercalcaemia.

4. Evaluation of a destructive bone lesion in a patient over 40

Rougraff diagnostic strategy (Rougraff et al., JBJS 1993): patient with a skeletal metastasis of unknown origin

1. **History** (smoking, weight loss, previous cancer, haematuria, breast lump) and **examination** (breast, prostate, thyroid, abdomen, lymph

nodes, skin).

2. **Laboratory:** FBC, ESR, **calcium**, **alkaline phosphatase**, renal and liver function, **serum and urine protein electrophoresis/free light chains (myeloma)**, **PSA** (men), thyroid function and others as indicated. Tumour markers (CEA, CA 15-3, CA-125) are of limited value.
3. **Plain radiographs** of the affected bone (whole bone) and **chest radiograph**.
4. **CT of chest, abdomen and pelvis**.
5. **Whole-body bone imaging: technetium-99m bone scan** (or PET-CT) to find other lesions and the most accessible site for biopsy.
6. **Biopsy** of the most accessible lesion (core needle, image-guided).
7. This approach **identified the primary in about 85%** of cases. **Lung (non-small cell) carcinoma was the commonest unknown primary**, followed by **kidney**. In about 10-15% no primary is found ("carcinoma of unknown primary").

> Exam pearl: Never fix a solitary destructive bone lesion until it is proven to be a metastasis. A "solitary metastasis" may be a primary bone sarcoma (needs wide resection, and intramedullary nailing would contaminate the whole bone), myeloma or renal carcinoma (needs embolisation). Biopsy first, following sarcoma biopsy principles; even with a known primary cancer, a first and solitary bone lesion should be biopsied.

Imaging details

- **Bone scintigraphy (Tc-99m MDP)**: sensitive whole-skeleton survey; **false negatives in purely lytic lesions** (**myeloma**, some **renal and thyroid** metastases, aggressive lytic lesions) and LCH; "**superscan**" (diffuse uptake with absent renal activity) in widespread prostate/breast metastases.

- **PET-CT**: FDG-avid lytic metastases; **PSMA PET-CT** for prostate cancer staging.
- **MRI**: **the most sensitive for marrow and spinal disease**, epidural extension and cord compression; whole-spine MRI for suspected MSCC.
- **CT**: cortical destruction, fracture risk, surgical planning; CT-guided biopsy.

Spinal clues

- **Vertebral body is involved first**; the **pedicle is destroyed on the AP radiograph ("winking owl" sign)**, because the body and posterior elements are involved together. Myeloma often spares the pedicle early.
- **Disc space is preserved** (tumours rarely cross the disc), in contrast to **pyogenic infection**, which destroys the disc and both endplates; **TB** may spare the disc early and spread subligamentously.
- **Malignant versus osteoporotic vertebral collapse on MRI:**

Feature	Malignant	Osteoporotic
Marrow signal	Diffuse T1 low signal replacing the **whole body**, often **multiple non-contiguous levels**	Band-like oedema, **preserved normal fatty marrow** in part of the body
Posterior cortex	**Convex bulge**	Retropulsed bony fragment, angular
Pedicle / posterior elements	**Involved**	Spared
Epidural/paravertebral mass	Present	Absent or thin
Other	Diffusion restriction, enhancement	**Fluid sign**, intravertebral vacuum cleft

5. Predicting pathological fracture

Mirels scoring system (Mirels 1989), long bones

Variable	1 point	2 points	3 points
Site	Upper limb	Lower limb	**Peritrochanteric**
Pain	Mild	Moderate	**Functional (mechanical)**
Lesion type	Blastic	Mixed	**Lytic**
Size (fraction of bone diameter on radiograph)	Under one-third	One-third to two-thirds	**Over two-thirds**

Total score	Fracture risk	Recommendation
7 or less	About 4% (low)	**Radiotherapy and observation**
8	About 15%	**Clinical judgement**
9 or more	**About 33% or more**	**Prophylactic fixation**

- Mirels score has **high sensitivity but poor specificity** (many patients scoring 9 would not fracture) and is less reliable for the femur at the margins of the scale; **functional pain is the most important clinical variable**.
- **Other criteria for prophylactic fixation of the femur:**
  - **Harrington's criteria: cortical destruction of 50% or more**, **lesion of 2.5 cm or more in the proximal femur**, **pathological avulsion of the lesser trochanter**, and **persistent pain on weight bearing despite radiotherapy**.
  - **Axial cortical involvement over 30 mm** (or circumferential involvement over 50%) on radiographs: the predictor validated in the **Dutch Femoral Metastases study (Van der Linden et al., 2004)**; lesions below this can be treated with radiotherapy alone.

  - **CT-based structural rigidity analysis (CTRA)** and **finite element models** are more accurate but not yet widely available.
- **Expected survival** guides the decision: **orthopaedic surgery is generally considered when expected survival is more than about 6 weeks** (long bones) and **3 months** (spine) in classic teaching; tools such as **PATHFx**, **SPRING** and **Katagiri** scores estimate survival.
- **Healing of pathological fractures (Gainor and Buchert 1983): myeloma highest (about 65-70%)**, **renal about 45%**, **breast about 35-40%**, **lung essentially 0%**. Healing depends on survival over 6 months; radiotherapy does not prevent healing but chemotherapy and poor nutrition reduce it. **Implants must therefore be designed to last without bone healing.**

6. Principles of surgical management (long bones)

- **Aims:** pain relief, **immediate weight bearing and function**, nursing care, and a construct that **outlasts the patient** (single operation).
- **Multidisciplinary discussion** (oncologist, radiotherapist, orthopaedic oncologist; BOA guidance on metastatic bone disease recommends this for all impending and actual fractures), optimise calcium, haemoglobin, nutrition; **embolise hypervascular lesions** (renal, thyroid, some myeloma).
- **Image the whole bone** (and the other femur) before fixation to detect additional lesions that must be protected.
- **Principles of fixation:**
  - **Protect the whole bone**: e.g. **long cephalomedullary (reconstruction) nail** for the femur so the femoral neck and entire shaft are protected.
  - Use **load-bearing or load-sharing** constructs that allow **immediate full weight bearing**.
  - **Curettage of the lesion and cement augmentation (PMMA)** to fill defects and improve screw purchase (with plates or nails).

  - **Replace rather than fix** where periarticular bone is destroyed: fractures of the **femoral head and neck do not heal** reliably: use **cemented (long-stem) hemiarthroplasty or total hip arthroplasty**, not internal fixation.
  - **Endoprosthetic (tumour/megaprosthesis) replacement** for extensive metaphyseal or periarticular destruction, failed fixation, or **solitary renal/thyroid metastasis treated for cure**.
- **Postoperative radiotherapy** to the **whole operative field** (entire implant length), usually started **about 2 weeks after surgery** once the wound is healing: **Townsend et al. (1994)** showed **fewer second procedures and better function** with postoperative radiotherapy.
- **Cardiopulmonary complications: fat and tumour embolism** with intramedullary reaming and **cemented long stems** (bone cement implantation syndrome): vent the canal, lavage, careful cementing, communicate with anaesthesia.

Choice of procedure by site

Site	Preferred treatment
Femoral head/neck	**Cemented hemiarthroplasty or THA** (long stem if more distal lesions); acetabulum assessed
Intertrochanteric / subtrochanteric	**Long cephalomedullary nail** (plus cement); **proximal femoral replacement** if extensive destruction or failed fixation
Femoral shaft	**Locked intramedullary nail** (protect the neck with a reconstruction nail)
Distal femur	Locking plate plus cement, retrograde nail, or **distal femoral replacement**
Proximal humerus	Intramedullary nail, plate plus cement, or **proximal humeral endoprosthesis/hemiarthroplasty**
Humeral shaft	**Locked intramedullary nail** (or plate plus cement)

Site	Preferred treatment
Tibia	Intramedullary nail; distal lesions plate plus cement
Pelvis (non-weight-bearing ilium, pubic rami)	Radiotherapy, non-operative

Periacetabular metastases: Harrington classification

Class	Defect	Treatment
I	Lateral cortices, superior and medial walls **intact**	**Conventional cemented THA**
II	**Medial wall deficient**	THA with a **protrusio cage/ring or mesh** to transfer load to the rim
III	**Lateral cortex and superior dome (roof) deficient**	**Harrington procedure: Steinmann pins (or screws) driven from the iliac crest into the acetabular region**, reinforced with **cement**, plus cage and cemented cup
IV	**Solitary lesion treated for cure** (or pelvic discontinuity)	**En bloc resection** and reconstruction

- Newer minimally invasive options: **percutaneous acetabuloplasty** (cementoplasty), ablation with screw fixation and cement.

7. Metastatic spine disease

Assessment framework: NOMS (Memorial Sloan Kettering)

- **N - Neurological:** degree of **epidural spinal cord compression (ESCC, Bilsky grade)** and myelopathy/radiculopathy.
- **O - Oncological: radiosensitivity** of the tumour.
- **M - Mechanical:** instability (**SINS**).

- **S - Systemic:** extent of disease, comorbidity, ability to tolerate surgery, expected survival.

Bilsky (ESCC) grading on axial T2 MRI

Grade	Description
0	Bone-only disease
1a	Epidural impingement **without deformation of the thecal sac**
1b	Thecal sac deformed, **no spinal cord abutment**
1c	Thecal sac deformed, **spinal cord abutted but not compressed**
2	**Cord compression, CSF still visible** around the cord
3	**Cord compression, no CSF visible**

- **Grades 0-1 = low grade**: radiotherapy (or SBRT); **grades 2-3 = high grade**: decompression is considered for radioresistant tumours.

Radiosensitivity

Radiosensitive	Radioresistant (to conventional external-beam radiotherapy)
Lymphoma, myeloma/plasmacytoma, seminoma, small cell lung, **breast**, **prostate** (most), Ewing	**Renal cell, thyroid, melanoma, sarcoma**, non-small cell lung, colon, hepatocellular

- **Stereotactic body radiotherapy (SBRT)** delivers high doses accurately and overcomes radioresistance; "**separation surgery**" (posterolateral decompression creating a 2-3 mm gap between tumour and cord, plus stabilisation) followed by SBRT is the MSKCC approach for high-grade ESCC of radioresistant tumours.

Spinal Instability Neoplastic Score (SINS; Spine Oncology Study Group, 2010)

Component	Scores
Location	**Junctional** (occiput-C2, C7-T2, T11-L1, L5-S1) **3**; mobile (C3-C6, L2-L4) 2; semi-rigid (T3-T10) 1; rigid (S2-S5) 0
Pain	**Mechanical pain 3**; occasional non-mechanical pain 1; pain-free 0
Bone lesion	Lytic 2; mixed 1; blastic 0
Radiographic alignment	**Subluxation/translation 4**; de novo deformity (kyphosis/scoliosis) 2; normal 0
Vertebral body collapse	Over 50% collapse 3; under 50% collapse 2; no collapse with over 50% body involved 1; none of these 0
Posterolateral involvement (facet, pedicle, costovertebral joint)	Bilateral 3; unilateral 1; none 0

- **Total 0-6 = stable**; **7-12 = indeterminate (potentially unstable)**; **13-18 = unstable**. **Surgical consultation is recommended for a score of 7 or more.**

Survival scores used to choose surgery

Score	Components	Interpretation
Revised Tokuhashi (0-15)	General condition (Karnofsky), extraspinal bone metastases, vertebral metastases, visceral metastases, primary site, neurological deficit	**0-8: under 6 months** (conservative/palliative); **9-11: 6 months or more** (palliative surgery); **12-15: 1 year or more** (excisional surgery)

Score	Components	Interpretation
Tomita (2-10)	Primary tumour growth rate (slow 1, moderate 2, rapid 4), visceral metastases (none 0, treatable 2, untreatable 4), bone metastases (solitary 1, multiple 2)	**2-3: wide or marginal excision (en bloc, long-term control)**; **4-5: marginal or intralesional excision**; **6-7: palliative surgery**; **8-10: supportive care**

Treatment of metastatic spinal cord compression

- **Immediate: dexamethasone** (e.g. 16 mg/day; higher bolus doses debated), analgesia, flat bed rest and spinal precautions if instability is suspected; **urgent whole-spine MRI (UK NICE: within 24 hours)**; spinal oncology and clinical oncology referral.
- **Patchell et al. (Lancet 2005) randomised trial: direct decompressive surgery (circumferential decompression and stabilisation) plus radiotherapy** versus **radiotherapy alone** for MSCC from a solid tumour:
  - **84% vs 57% able to walk after treatment**; **non-ambulant patients regained walking in 62% vs 19%**; walking retained longer (median 122 vs 13 days); less steroid and opioid use; trial stopped early for benefit.
  - **Inclusion:** single area of compression, **expected survival 3 months or more**, **paraplegia for less than 48 hours**, **not radiosensitive tumours** (lymphoma, myeloma, leukaemia, germ-cell excluded).
- **Radiotherapy alone:** radiosensitive tumours, poor surgical candidates, multilevel disease, long-standing complete paraplegia (over 48 hours), short life expectancy.
- **Surgical options:** posterior decompression and **instrumented stabilisation** (laminectomy alone is contraindicated when the anterior column is involved, as it worsens instability), separation

surgery, corpectomy and reconstruction, **en bloc spondylectomy (Tomita)** for solitary renal or thyroid metastasis with good prognosis; **minimally invasive percutaneous screw fixation**.

- **Vertebral augmentation (vertebroplasty or kyphoplasty)** for painful pathological compression fractures without significant epidural disease or instability: **CAFE trial (Berenson et al., Lancet Oncology 2011)**: **balloon kyphoplasty gave rapid and sustained improvement in pain and function** compared with non-surgical care. Cement leakage is the commonest complication.

8. Non-surgical treatment

Radiotherapy

- Pain relief in about **60-70%** of patients, often within 2-4 weeks.
- **Single fraction 8 Gy gives the same pain relief as multiple fractions** (e.g. 30 Gy in 10, 20 Gy in 5), confirmed by **RTOG 9714** and the **Dutch Bone Metastasis Study**; **retreatment rate is higher after single fraction (about 18-20% vs 8-10%)**. Single fraction is preferred for uncomplicated painful metastases and poor prognosis.
- **Stereotactic (SBRT)** for oligometastases, radioresistant tumours and re-irradiation; risk of **vertebral compression fracture** after SBRT.
- **Radiopharmaceuticals: radium-223** (alpha emitter; **ALSYMPCA trial**: improved overall survival in castration-resistant prostate cancer with bone metastases), strontium-89 and samarium-153 (palliation, marrow suppression).

Bone-targeted agents

- **Bisphosphonates (zoledronic acid):** inhibit osteoclasts (**farnesyl pyrophosphate synthase** in the mevalonate pathway); **reduce skeletal-related events** in breast, prostate, lung cancer and myeloma; treat hypercalcaemia.
- **Denosumab (monoclonal antibody against RANKL): superior to zoledronic acid in delaying the first SRE** in breast (Stopeck et al.,

2010) and prostate cancer (Fizazi et al., 2011); no renal dose adjustment but risk of **hypocalcaemia** (give calcium and vitamin D); rebound bone loss on stopping.

- **Complications: osteonecrosis of the jaw** (dental assessment before starting), **atypical femoral fractures** (lateral cortical beaking: watch for thigh pain), acute-phase reaction, renal toxicity (zoledronic acid).
- **Systemic oncological therapy** (hormonal therapy for breast and prostate, targeted agents, immunotherapy) controls disease and improves healing.
- **Analgesia** (WHO ladder), physiotherapy, orthoses, palliative care.

9. Special situations

- **Solitary renal cell carcinoma metastasis: wide (en bloc) excision** may **improve survival** (especially metachronous solitary lesions after nephrectomy with a long disease-free interval); **embolise preoperatively**; renal metastases respond poorly to radiotherapy and fixation alone often fails (progression around the implant), so **resection and endoprosthetic reconstruction** is favoured.
- **Thyroid metastases:** radioiodine-avid lesions may respond to **radioactive iodine**; hypervascular; solitary lesions resected.
- **Prostate:** blastic lesions are painful but fracture less often; respond to androgen deprivation; PSMA-targeted therapy (lutetium-177 PSMA).
- **Breast:** mixed lesions, hormone-responsive; long survival possible, so durable reconstructions are justified.
- **Lung:** shortest survival (often months), fractures rarely heal: simplest effective procedure.
- **Impending fracture versus completed fracture: prophylactic fixation** of an impending fracture gives **shorter hospital stay, less blood loss, faster return to walking and better survival** than

fixation after the fracture (retrospective data); hence treat impending fractures early.

10. Differential diagnosis of a lytic lesion in an older adult (distractors)

Condition	Clue
Myeloma	Punched-out lesions, normal ALP, cold bone scan, M-protein, anaemia, renal failure
Primary bone sarcoma (chondrosarcoma, UPS of bone, osteosarcoma, secondary sarcoma in Paget disease)	Solitary lesion, matrix calcification, periosteal reaction, soft-tissue mass; **biopsy before fixation**
Lymphoma	Permeative, large soft-tissue mass, well patient
Brown tumour of hyperparathyroidism	High calcium, high PTH, low phosphate, subperiosteal resorption of phalanges
Paget disease	Coarse trabeculae, cortical thickening, enlarged bone, raised ALP; **osteoporosis circumscripta** of skull in lytic phase
Osteoporotic insufficiency fracture	Typical sites (sacrum H-sign, femoral neck), MRI signal pattern
Infection (pyogenic, tuberculosis)	Disc involvement in spine, fever, raised CRP
Giant cell tumour	Epiphyseal-metaphyseal, eccentric, skeletally mature (20-40 years)

Exam pearl: Breast, prostate, lung, kidney and thyroid cause most bone metastases; spine is the commonest site and proximal femur the commonest long bone. Mirels 9 or more means prophylactic fixation. Fix the whole bone, use cement, replace (do not fix) the femoral neck, give postoperative radiotherapy to the whole implant, and embolise renal and thyroid metastases. For spinal cord compression from a radioresistant solid tumour, decompression with stabilisation plus radiotherapy beats radiotherapy alone (Patchell 2005).

Written by Claude for Ortho preSSaleel · updated 2026-11-15. Check doubtful points in a standard textbook.